

Juan Ferro

(Updated October 2025)

jj.ferro26 @ uniandes.edu.co

Fourth year PhD student in the economics department at Universidad de los Andes (Bogotá, Colombia). Interested in Applied microeconomics with research on Crime economics, Public Economics and the study of Social Norms.

WORKING PAPERS

Behavioral responses to fines: direct and indirect deterrence (Job market paper) (link)

Abstract: Governments use fines in order to deter socially unwanted behavior on a wide array of issues. It is commonly assumed that, fixing the probability of sanction, the higher the price, the more deterrent a fine is. But with limited government capacity for enforcement, the fine is often only an invitation for citizens to change behavior. Thus, the perceived fairness of the fine price might moderate deterrence. I explore this behavioral mechanism with data from automated speed cameras in Bogotá. Using the real speed limit as a source of exogenous variation and a regression discontinuity design, I find that on average drivers change their behavior (reduce speeding) where the law is enforced. But there is no average behavioral change where the law is not enforced (no change in road crashes). Using the commercial price of the vehicle as a proxy for wealth and the fact that in this context the monetary sanction is fixed, I can also measure the heterogeneous effect of relative price. I consistently show that the direct deterrence effect of a fine (speeding) is explained by higher wealth levels. The indirect deterrence effect (crashes) has a different relationship with wealth: fines increase crashes at lower levels (quintile 2) while reducing them at higher levels (quintile 5). The behavioral story, direct and indirect deterrence only for the wealthy with no direct deterrence and backfire on indirect deterrence for the less wealthy, is even more clear when only looking at two groups, above and below median price. My findings are consistent with fines being an invitation by the government to broadly change behavior that is only accepted if it is perceived to be fair. This has specific public policy implications: speeding tickets might not be a well targeted policy for preventing road fatalities. Contrary to extended belief, in some contexts lower fines might be a more behaviorally effective policy tool.

Making inequality unacceptable: an experiment on fiction reading and prosociality (with Jose Guerra)

With funding from III TREES Research Grant Fund

In this field experiment we test the widely accepted hypothesis that fiction reading increases prosociality. Participants are assigned randomly to a four week course on fiction reading or a four week course centered on insights from behavioral economics. We add three elements to this literature. First, a casual estimate from the randomization and construction of a placebo treatment. Second, the use of a revealed preference approach centered on economic games that complements existing self-reported measures from the psychological literature. Third, the measurement of short and medium term effect by capturing preference four weeks after the intervention.

Pilot experiment to be run at Universidad de los Andes in October and November 2025.

Police scandals and deterrence: an exploration from traffic offenses and accidents (link)

Abstract: Increasing the deterrent effect of sanctions is a pressing public policy issue in many areas where the state has limited capacity to punish all wrongdoers. But if institutions in charge of enforcing the law are perceived as illegitimate, those sanctions might be less deterrent. This paper uses local media, traffic tickets and road accidents in Bogotá, Colombia to study whether sanctions imposed by more legitimate authorities are more deterrent. My estimation compares the behavior of drivers that get their first tickets on days just after a police scandal with drivers that get their first tickets just before a scandal. Drivers that get their tickets just after a police scandal are more likely to not pay their fine and get a second ticket. Furthermore, they are 21% (11%) more likely to be in an accident in the following six months than drivers that get their first ticket just before a scandal (any other day). My results suggest that in this policy issue increasing the legitimacy of existing sanctions might be more effective than increasing the number of sanctions.

The role of legitimacy on deterrence: a speed-control intervention (link)

Abstract: The deterrence effect of legal sanctions has been widely studied and measured. Less attention has been given

to the mechanisms by which some sanctions are more deterrent than others. This paper suggests one possible mechanism: the legitimacy of legal sanctions. I present a model in which individuals respond to a legal sanction for previously non-regulated behavior. An implication of the model is that more legitimate sanctions, those with higher social acceptance of the mandated behavior, cause a larger change in the targeted behavior. I test this hypothesis on the *Cámaras Salvavidas* Program, a speed-control intervention in Bogotá, Colombia.

PERSONAL INFORMATION

Sex: male
Citizenship: Colombian

EDUCATION

Universidad de los Andes, Bogotá, Colombia

PhD in Economics, march 2026 (expected graduation)
Advisors: Jose Alberto Guerra and Ignacio Sarmiento Barbieri.

Master in Economics, 2020.
Research thesis: *The role of gender and sexuality stereotypes on teenage childbearing in Colombia.*
Advisors: Darío Maldonado, School of government, Universidad de los Andes.
Elective courses: Experimental economics, Impact evaluation: statistics and economics, The economics of cities.

Universidad de los Andes, Bogotá, Colombia
Bachelor of Arts, Political Science, 2012

Universidad de los Andes, Bogotá, Colombia
Bachelor of Arts, Law, 2011

TEACHING EXPERIENCE

Microeconomics 101 Spring 2025, 2026
(Full course)
Mandatory course for all undergraduate students in the Social Science department (Political Science, Psychology)

Game theory Summer 2025, Summer 2024
(Summer course instructor)
Mandatory course for all undergraduate students in economics

Advanced microeconomics (masters and Phd course) Fall 2024
(Teaching assistant to Paula Jaramillo and José Guerra)

Microeconomics 101 Spring 2024
(Teaching assistant to José Guerra and Andrés Moya)

Introductory microeconomics (master level course) Spring 2024
(Teaching assistant to Andrés Álvarez)

Advance econometrics (Phd course) Fall 2023
(Teaching assistant to Oscar Becerra)

OTHER EMPLOYMENT

Mobility Office of Bogotá

Consultant on Data Analysis and Policy

March 2024-

OTHER

Software and programming: Stata, R.

Languages:

- Spanish (native).
- English (excellent). TOEFL 112/120.
- French (basic knowledge)

Other interests: Literature. Author of the novel Experimental Economics (angosta editores, 2023), National Prize of Literature of Colombia for unpublished novels in 2022.

CONTACT INFORMATION

Email: jj.ferro26@uniandes.edu.co
Office W-717, Universidad de los Andes,
Cra 1 N° 18A – 12, Bogotá, Colombia
Phone: (+57) 3123144097